

Characterize Markov perfect equilibrium under unobservable inspection times in opportunity hunting

ChatGPT (gpt-5.6-sol)*

Partial progress generated on 2026-08-24

Abstract

We present partial progress on an open problem from *Opportunity Hunters: A Model of Competitive Sequential Inspections*.

1 The open problem

This paper addresses an open problem posed by Ran Eilat, Zvika Neeman, Eilon Solan in *Opportunity Hunters: A Model of Competitive Sequential Inspections* [1] (Economics and Computation (EC), 2023), Section 6.2.1 (Observable vs. Unobservable Actions), page 29.. The website catalogue entry is problem 3 (W4383533306_p0) of the [Open Problems in Operations Research](#) collection.

In the opportunity-hunting game described above with parameters $(\lambda, r, c, v_1, v_2)$, assume inspection times are not observable to the other player (private actions), while discovery of the prize ends the game publicly.

Characterize the Markov perfect equilibrium (existence, and when it exists, the equilibrium strategy profile and induced equilibrium payoff/inspection rate) under this unobservable-actions information structure, in the parameter regime where the follower prize is large (i.e., the regime in which the observable-actions model yields a mixed, continuously supported equilibrium inspection-time distribution).

The remainder of this document is the partial progress produced by the model, reproduced verbatim from the pipeline output.

Status: partial progress, not a complete solution of the unrestricted private-action MPE problem. The designated-inspector equilibria can be completed rigorously, simultaneous inspections can be handled under the canonical tie rule, and the deterministic classifications can be stated correctly. The mixed equilibrium for $v_2^{\text{pub}} < v_2 < \bar{v}_2$, and the full mixed-equilibrium set for larger v_2 , remain unresolved.

*Generated by the `base_solver_ni_two_iterations` pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve $\rightarrow$ A.8 verify $\rightarrow$ A.10 repair write-up; run `run_20260824_024352_98dd0fab, ec-2it-ultra`); two-iteration run, outcome from iteration 2. Independent verifier assessment: REJECTED with progress score 2/3 (major partial progress). Maintained by the AI in Mathematics & Operations Research project.

Assume $r, \lambda > 0$, $0 < c < v_1$, locally finite inspection schedules, and that simultaneous inspectors each pay c and are assigned leader and follower with equal probability. Put

$$\kappa = r + \lambda, \quad B = v_1 - c,$$

and

$$g(t) = e^{-rt}((1 - e^{-\lambda t})v_1 - c) = Be^{-rt} - v_1e^{-\kappa t}.$$

Define

$$w(t) = \frac{g(t)}{1 - e^{-\kappa t}}.$$

The sign of $w'(t)$ is the sign of

$$\kappa v_1 - rBe^{\lambda t} - \lambda Be^{-rt}.$$

This expression is strictly decreasing from $\kappa c > 0$ to $-\infty$. Hence w has a unique maximizer $h > 0$, satisfying

$$\kappa v_1 e^{-\lambda h} = B \left(r + \lambda e^{-\kappa h} \right), \quad (1)$$

and

$$w^* := w(h) = \frac{\lambda B}{\kappa} e^{-rh}. \quad (2)$$

Set

$$v_2^{\text{pub}} = v_1 - \frac{c}{1 - e^{-\lambda h}}$$

and

$$\bar{v}_2 = \max_{0 < t \leq h} e^{r(h-t)} \left(v_1 - \frac{c}{1 - e^{-\lambda t}} \right)_+. \quad (3)$$

As in the rejected attempt,

$$0 < v_2^{\text{pub}} < \bar{v}_2 < B. \quad (4)$$

The strict inequality follows because the derivative at h of

$$e^{r(h-t)} \left(v_1 - \frac{c}{1 - e^{-\lambda t}} \right)$$

is negative. Its maximizer $\hat{t} < h$ is determined by

$$rv_1 p^2 + c(\lambda - r)p - c\lambda = 0, \quad p = 1 - e^{-\lambda \hat{t}}. \quad (5)$$

1. Simultaneous deterministic inspections are not an equilibrium issue

The earlier attempt excluded ties. Under the canonical rule they can instead be ruled out in every deterministic equilibrium whenever $v_1 > v_2$, which includes the range $v_2 \leq B$.

Suppose that, following a commonly known absence, both players are scheduled to inspect simultaneously after a gap $x > 0$. Let C be the continuation payoff following a joint failure. A player's prescribed conditional payoff is

$$U^{\text{tie}} = e^{-rx} \left[(1 - e^{-\lambda x}) \frac{v_1 + v_2}{2} - c + e^{-\lambda x} C \right].$$

Instead, that player can inspect at $x - \varepsilon$, skip the inspection at x after a failure, and resume the prescribed continuation after the opponent's inspection at x . As $\varepsilon \downarrow 0$,

$$U^{\text{preempt}} - U^{\text{tie}} \longrightarrow \frac{1}{2}e^{-rx}(1 - e^{-\lambda x})(v_1 - v_2) > 0. \quad (6)$$

The cost and continuation terms converge to the same quantities; the strict gain comes from becoming the sole leader rather than receiving the average tie prize. Therefore:

Tie lemma. If $v_1 > v_2$, no deterministic equilibrium has an on-path simultaneous inspection.

Thus the subsequent deterministic no-tie analysis applies to the canonical game rather than to an artificially restricted version.

2. Fully completed designated-inspector equilibria

Consider a profile in which player A is active and player P is passive.

The active player's strategy after a failed inspection is to inspect again after h . At an off-path state where the last known absence was a units ago, the active player waits $h - a$ if $a < h$, and inspects immediately if $a \geq h$. The passive player never inspects.

Active player's arbitrary deviations

Against a permanently passive opponent, any pure all-failure inspection schedule has gaps $X_1, X_2, \dots$, with $S_n = X_1 + \dots + X_n$. Its expected payoff is

$$\sum_{n \geq 1} e^{-\kappa S_{n-1}} g(X_n).$$

Because

$$g(x) \leq w^*(1 - e^{-\kappa x}), \quad (7)$$

partial sums telescope:

$$\sum_{n=1}^N e^{-\kappa S_{n-1}} g(X_n) \leq w^*(1 - e^{-\kappa S_N}) \leq w^*.$$

Local finiteness implies $S_N \rightarrow \infty$ for an infinite schedule, so the terminal term vanishes. The same inequality holds after averaging over randomized or history-dependent schedules. Equality requires every gap to equal h .

For an off-path elapsed age a , the value from making the next inspection at total age $t \geq a$ is

$$g(t) + e^{-\kappa t} w^* = B e^{-rt} - (v_1 - w^*) e^{-\kappa t}.$$

This is increasing up to h and decreasing thereafter, proving the stated off-path policy.

Passive player's arbitrary deviations

Let $H \in (0, h]$ be the time from a known absence to the active player's next scheduled inspection. If the passive player never inspects, its value is

$$W(H) = e^{-rH} \left[(1 - e^{-\lambda H}) v_2 + e^{-\lambda H} W(h) \right], \quad (8)$$

where

$$W(h) = \frac{e^{-rh}(1 - e^{-\lambda h})v_2}{1 - e^{-\kappa h}}. \quad (9)$$

Inserting an inspection at $t < H$, followed after failure by returning to passivity, changes the payoff by

$$\Delta(t, H) = (1 - e^{-\lambda t}) \left[e^{-rt} \left(v_1 - \frac{c}{1 - e^{-\lambda t}} \right) - e^{-rH} v_2 \right]. \quad (10)$$

If $v_2 \geq \bar{v}_2$, then for every $0 < t < H \leq h$,

$$e^{r(H-t)} \left(v_1 - \frac{c}{1 - e^{-\lambda t}} \right)_+ \leq \bar{v}_2 \leq v_2,$$

so $\Delta(t, H) \leq 0$.

Inspecting exactly at H , and hence tying with the active player, changes the payoff by

$$\Delta_{\text{tie}}(H) = e^{-rH} \left[\frac{1 - e^{-\lambda H}}{2} (v_1 - v_2) - c \right]. \quad (11)$$

Since $v_2 \geq \bar{v}_2 > v_2^{\text{pub}}$,

$$v_1 - v_2 < \frac{c}{1 - e^{-\lambda h}},$$

and hence $\Delta_{\text{tie}}(H) < 0$ for every $H \leq h$.

These one-inspection comparisons rule out arbitrary deviations as follows. In any active-player interval, local finiteness gives a last deviating inspection. Delete that inspection conditional on the preceding failed inspection; (10), or (11) if it is a tie, shows that deletion weakly increases payoff. Repeating backwards deletes every deviation in that interval. Apply this through the first N active intervals. The positive part of any continuation payoff is bounded by $\max\{v_1, v_2\}$, so the discounted remainder is at most

$$e^{-rNh} \max\{v_1, v_2\} \rightarrow 0.$$

This supplies the missing backward-deletion and transversality argument.

Conversely, if $v_2 < \bar{v}_2$, choosing $\hat{t}$ in the first length- h interval makes (10) strictly positive. Therefore:

Designated-inspector theorem. A designated-inspector equilibrium exists if and only if

$$v_2 \geq \bar{v}_2. \quad (12)$$

There are two such role assignments. Their payoffs are

$$V_A = w^*, \quad V_P = \frac{e^{-rh}(1 - e^{-\lambda h})v_2}{1 - e^{-\kappa h}}. \quad (13)$$

Their all-failure scheduled inspection rates are

$$\nu_A = \frac{1}{h}, \quad \nu_P = 0. \quad (14)$$

The active player's expected undiscounted number of inspections before discovery is

$$\mathbb{E}N_A = \frac{1}{1 - e^{-\lambda h}},$$

and the passive player's is zero.

The assessment is sequentially complete. After each scheduled active inspection with no discovery, the limiting Bayesian belief assigns probability one to a failure and therefore to absence at that instant. After any private off-grid failure, the inspector knows absence at that time. The passive player's payoff-relevant state is the time since its latest private or inferred absence and the time until the next active inspection; the proof above applies for every such state. These beliefs arise as limits of full-support trembles. Thus the designated profiles are belief-Markov sequential equilibria, not merely on-path calendars.

3. Deterministic impossibility below $\bar{v}_2$

By the tie lemma it suffices to consider a deterministic aggregate calendar

$$0 = s_0 < s_1 < s_2 < \dots, \quad x_n = s_n - s_{n-1},$$

with a unique owner o_n at every s_n .

If $v_2 < \bar{v}_2$, every gap satisfies

$$x_n < h. \tag{15}$$

Indeed, if $x_n \geq h$, the player not inspecting at s_n can inspect at $\hat{t}$ after s_{n-1} and then revert after failure. By (10), absence of a gain would require

$$v_2 \geq e^{r(x_n-h)} \bar{v}_2 \geq \bar{v}_2,$$

a contradiction. At $v_2 = \bar{v}_2$, the same argument yields $x_n \leq h$.

A small perturbation of s_n , holding s_{n-1}, s_{n+1} fixed, gives the following necessary first-order conditions. If $o_{n+1} = o_n$,

$$\kappa v_1 = rB e^{\lambda x_n} + \lambda B e^{-r x_{n+1}}. \tag{16}$$

If ownership switches,

$$\kappa v_1 = rB e^{\lambda x_n} + \lambda v_2 e^{-r x_{n+1}}. \tag{17}$$

Define

$$D(x) = \kappa v_1 - rB e^{\lambda x} - \lambda B e^{-r x}. \tag{18}$$

Then $D(h) = 0$ and

$$D'(x) = r\lambda B(e^{-rx} - e^{\lambda x}) < 0.$$

Hence $D(x) > 0$ for $x < h$.

Equation (16) implies $x_{n+1} < x_n$. Equation (17), using $v_2 < B$, implies the same strict inequality. Thus, if any gap is below h ,

$$x_n \downarrow \ell < h.$$

Along an infinite subsequence of same-owner transitions, (16) would give $D(\ell) = 0$. Along an infinite subsequence of switches, (17) would give

$$\kappa v_1 = rB e^{\lambda \ell} + \lambda v_2 e^{-r \ell} < rB e^{\lambda \ell} + \lambda B e^{-r \ell} < \kappa v_1,$$

also impossible.

A finite schedule is impossible because, following its last failure, a sufficiently delayed single inspection has positive payoff $g(t) > 0$.

It follows that:

- If $v_2 < \bar{v}_2$, there is no locally finite deterministic equilibrium in the canonical game.
- If $v_2 = \bar{v}_2$, every gap must equal h . A switch would make (17) imply

$$e^{-rx_{n+1}} = \frac{B}{\bar{v}_2} e^{-rh},$$

hence $x_{n+1} < h$, a contradiction. Thus the only deterministic on-path calendars are the two designated-inspector calendars. Section 2 supplies their sequential completions.

This does not exclude mixed equilibria at or below $\bar{v}_2$.

4. The endpoint $v_2 = B$

At $v_2 = B$, deterministic ties remain impossible by (6). Equations (16) and (17) coincide:

$$\kappa v_1 = rBe^{\lambda x_n} + \lambda Be^{-rx_{n+1}}. \quad (19)$$

The probe deviation gives the uniform bound

$$x_n \leq h + \frac{1}{r} \log \frac{B}{\bar{v}_2}.$$

If $x_n < h$, (19) implies $x_{n+1} < x_n$; if $x_n > h$, it implies $x_{n+1} > x_n$. Boundedness and the uniqueness of the fixed point h rule out either alternative. Thus every aggregate gap equals h .

Conversely, assign every point of

$$h, 2h, 3h, \dots \quad (20)$$

to exactly one player. To verify best responses, consider two consecutive opponent inspections separated by $P = mh$. Suppose the player makes inspections with preceding gaps $\ell_1, \dots, \ell_k$, leaving a final gap u before the opponent inspection. Monopoly optimality gives

$$g(\ell) \leq w^*(1 - e^{-\kappa\ell}).$$

Consequently its payoff in that opponent interval, omitting the common continuation after the opponent's failure, is at most

$$w^* + e^{-\kappa(P-u)}(L_B(u) - w^*), \quad L_B(u) = B(e^{-ru} - e^{-\kappa u}). \quad (21)$$

Maximizing the last term is equivalent to maximizing

$$K_B(u) = B(e^{\lambda u} - 1) - w^* e^{\kappa u}.$$

Using (2),

$$K'_B(u) = \lambda B e^{\lambda u} (1 - e^{r(u-h)}),$$

so $u = h$ is the unique maximizer. Equality in (21) also requires every preceding gap to equal h . Hence the unique interval best response is to inspect at all intermediate grid points not assigned to the opponent. Tying at the opponent's endpoint is strictly worse than waiting because

$$\frac{1 - e^{-\lambda u}}{2}(v_1 - B) - c = \frac{c}{2}(1 - e^{-\lambda u}) - c < 0.$$

Backward interval deletion and discounted transversality now show that no randomized multiple-inspection deviation improves the payoff.

An off-path sequential completion is obtained as follows: after a private off-grid failure, each player assigns probability one to the opponent continuing its deterministic assigned calendar and uses an optimal policy for that deterministic-deadline impulse problem. Finite-horizon backward induction followed by the discounted limit gives such an optimal policy; the lower bound on inspection gaps prevents explosion. Therefore every coloring admits a sequential-equilibrium completion. A nonperiodic coloring is not thereby shown to be a stationary MPE: its continuation may depend on the entire residual calendar. Periodic colorings are belief-Markov only if the public phase is admitted as part of the Markov state.

Let

$$d = e^{-rh}, \quad q = e^{-\lambda h}, \quad \rho = dq, \quad F = \frac{d(1-q)B}{1-\rho}.$$

For a coloring $\sigma_n \in \{1, 2\}$,

$$U_i = F - c \sum_{n: \sigma_n = i} \rho^n. \quad (22)$$

Thus the exact deterministic payoff set is

$$\left\{ \left(F - cx, F - c \left(\frac{\rho}{1-\rho} - x \right) \right) : x = \sum_{n \in A} \rho^n, A \subseteq \mathbb{N} \right\}. \quad (23)$$

If the coloring has an asymptotic ownership fraction θ , its scheduled rates are

$$\frac{\theta}{h}, \quad \frac{1-\theta}{h}. \quad (24)$$

For colorings without an asymptotic density, ordinary long-run individual rates need not exist.

These statements classify deterministic on-path calendars and their sequential completions at $v_2 = B$; they do not classify mixed MPE.

5. Correct belief-state best-response system for mixed strategies

The earlier equations omitted $t = \infty$, mishandled atoms, and did not define a forward operator. The following is the corrected reduction. It is not an existence theorem.

Consider a private failure at time s . Let μ be the player's posterior law of the opponent's complete locally finite all-failure inspection configuration ω . If the arrival time is $A > s$, define the opponent's discovery time

$$D(\omega, A) = \inf\{u \in N_\omega : u \geq A\},$$

with $D = \infty$ if no such inspection exists.

For $t > s$, put

$$S_\mu^-(t) = \Pr_\mu(D \geq t), \quad S_\mu^+(t) = \Pr_\mu(D > t).$$

Thus

$$S_\mu^-(t) - S_\mu^+(t) = \Pr_\mu(D = t)$$

is the opponent-discovery atom at t . Writing $x = t - s$,

$$S_\mu^+(t) = e^{-\lambda x} + \int_s^t \lambda e^{-\lambda(a-s)} \mu(N_\omega \cap [a, t] = \emptyset) da, \quad (25)$$

while

$$S_\mu^-(t) = e^{-\lambda x} + \int_s^t \lambda e^{-\lambda(a-s)} \mu(N_\omega \cap [a, t] = \emptyset) da. \quad (26)$$

Let

$$z = \frac{v_1 + v_2}{2}.$$

If the player chooses its next inspection at finite t , its exact value is

$$\begin{aligned} J_{i,s,\mu}(t) = & v_2 \int_{[s,t)} e^{-r(u-s)} \Pr_\mu(D \in du) \\ & + e^{-rx} \left[v_1 (S_\mu^+(t) - e^{-\lambda x}) + z (S_\mu^-(t) - S_\mu^+(t)) \right. \\ & \left. - c S_\mu^-(t) + e^{-\lambda x} V_i(t, R_t \mu) \right]. \end{aligned} \quad (27)$$

The terms respectively represent:

- becoming follower before t ;
- becoming sole leader at t ;
- tying an opponent discovery atom at t ;
- paying the inspection cost whenever the game has not ended strictly before t ;
- continuation after a failed inspection.

Here $R_t \mu$ is the push-forward of μ to the opponent's residual hidden state at t . Conditional on one's own failure, $A > t$; because the arrival process is independent of the opponent's private random seed, no additional likelihood tilt of μ is needed.

The missing passive action is

$$J_{i,s,\mu}(\infty) = v_2 \int_{[s,\infty)} e^{-r(u-s)} \Pr_\mu(D \in du). \quad (28)$$

Define

$$\eta = -\frac{1}{\lambda} \log \left(1 - \frac{c}{v_1} \right). \quad (29)$$

The next-action Bellman equation must therefore use the compactified domain

$$V_i(s, \mu) = \sup_{t \in [s+\eta, \infty]} J_{i,s,\mu}(t). \quad (30)$$

Bayesian filtering while waiting is explicit. For example, after observing no discovery through t , the likelihood of an opponent path ω is

$$L_t^+(\omega) = e^{-\lambda x} + \int_s^t \lambda e^{-\lambda(a-s)} \mathbf{1}\{N_\omega \cap [a, t] = \emptyset\} da,$$

and hence

$$\mathcal{B}_t^+ \mu(d\omega) = \frac{L_t^+(\omega) \mu(d\omega)}{\int L_t^+ d\mu}. \quad (31)$$

The left-limit filter replaces $[a, t]$ by $[a, t)$.

For measurable next-inspection kernels $\sigma_i(s, \mu, \cdot)$, let $\Gamma_i(\sigma_i; \mathcal{B}_i(P_j))$ denote the all-failure path law generated recursively by drawing the next inspection from σ_i , applying the failure update R_t , and repeating. A belief-consistent mixed MPE must solve

$$\begin{cases} P_i = \Gamma_i(\sigma_i; \mathcal{B}_i(P_j)), \\ \mu_i(h_i) = \mathcal{B}_i(P_j \mid h_i), \\ \sigma_i(s, \mu, \operatorname{argmax}_{t \in [s+\eta, \infty]} J_{i,s,\mu}(t)) = 1, \end{cases} \quad i = 1, 2. \quad (32)$$

Off-path beliefs must additionally be limits of full-support perturbations.

If bounded measurable kernels satisfying (32) exist, backward optimization plus discounted transversality makes (32) sufficient for sequential optimality. What is not proved is that the argmax is always nonempty or admits a measurable selector, that the induced Bayesian operator is continuous at atoms, or that (32) has a fixed point. Those are precisely parts of the unresolved existence problem.

6. What the posterior and refractory calculations actually prove

Suppose the current probability that the opportunity is present is p , and an opponent privately inspects with probability α . An inspector who fails has posterior

$$p^F = 0,$$

whereas a noninspector observing no public discovery has

$$p^N = \frac{p(1-\alpha)}{1-\alpha p} > 0. \quad (33)$$

Thus an unobserved mixed inspection does not generate a common public reset. This does not prove that every equilibrium requires an irreducibly infinite-dimensional state; a special equilibrium might have a finite-dimensional sufficient statistic.

After a known absence, conditional on no opponent discovery during an interval of length x , the probability of arrival is no greater than $1 - e^{-\lambda x}$. Deleting an inspection and simulating a failure therefore loses at most

$$(1 - e^{-\lambda x})v_1$$

while saving c . Consequently every inspection with $x < \eta$ is strictly dominated, and one at $x = \eta$ is only weakly dominated. Hence every player's all-failure schedule has gaps at least η and upper asymptotic rate at most $1/\eta$.

This rules out an unconditioned translation-invariant positive-intensity process initialized directly at a known-absence reset. It does not rule out a time-homogeneous age-dependent policy, a delayed renewal process, or a stationary Palm regime after an initial delay.

Final regime statement

For the canonical random tie rule and $v_2 \leq B$, the rigorously established conclusions are

$v_2^{\text{pub}} < v_2 < \bar{v}_2$	no deterministic equilibrium; mixed existence/payoffs/rates open
$v_2 = \bar{v}_2$	exactly two deterministic on-path calendars, both designated; mixed set open
$\bar{v}_2 < v_2 < B$	at least two completed designated-inspector MPE; full equilibrium set open
$v_2 = B$	all deterministic on-path calendars are h -grid colorings; they admit sequential completions, but mixed MPE remain unclassified.

Thus the designated-inspector construction, its off-path completion, the arbitrary-deviation proof, and the deterministic tie issue are repaired. The central mixed private-action MPE is not solved, and unrestricted equilibrium characterization remains open throughout the large- v_2 regime.

References

- [1] Ran Eilat, Zvika Neeman, Eilon Solan. *Opportunity Hunters: A Model of Competitive Sequential Inspections*. Economics and Computation (EC), 2023. <https://doi.org/10.1145/3580507.3597760>